DO NOW — RECALL

Timed Paper 2 algorithms practice

Name: _____ Class: _____ Date: _____

1. Which structure retrieves the most recently added item first?

2. What is the sorted-input precondition?

3. What moves right as a key is inserted into a sorted prefix?

4. What are the worst-case time complexities of linear and binary searches?

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

Stack (LIFO).

2. Model answer

The array must be sorted.

3. Model answer

All preceding values greater than the key.

4. Model answer

Linear O(n); binary O(log n) for sorted input.

Vocabulary — know these terms

trace — Record an algorithm's changing state.

sorted prefix — Part already in order during insertion sort.

precondition — Requirement, such as ordering, for an algorithm.

O(log n) — Work grows by repeated halving.

auxiliary space — Extra storage required by an algorithm.

NEW LEARNING

Read, annotate and explain

Read the task command

A complete pass means all permitted adjacent comparisons for that pass, not just one swap.

Trace one pass

[6,2,5,1] becomes [2,5,1,6] after comparing each pair.

Finish every pass

Continue to [2,1,5,6], then [1,2,5,6]. A clear trace shows each pass.

Explain one step in your own words, using the correct technical terms.

I DO — WORKED EXAMPLE

Follow the worked method

Model the first complete ascending bubble-sort pass on [4,1,3].

Predict first

Start [4,1,3] and compare 4 with 1: swap.

Check the method

Now compare 4 with 3: swap, producing [1,3,4].

Check the method

The pass is complete and the array is sorted.

WE DO — GUIDED PRACTICE

Try → Discuss → Improve

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 1

An array contains [8,3,6,1]. Trace ascending insertion sort, giving the array after each outer-loop iteration and explaining how the key is positioned.

Question 2

Write a C# function BinarySearch(int[] a,int target) returning the first matching index it encounters, or −1. Include correct bounds and a sorted-input precondition.

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 3

Explain one advantage and one disadvantage of merge sort over bubble sort, including time complexity, and identify an input scenario for which linear search is more practical than binary search.

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking the answer.

Binary search works correctly on any unordered input.

How confident are you? Circle one: Confident / Getting there / Need help

Explain your choice, identifying one concept you can now use independently.

GREEN PEN — Correct and explain

Choose one mistake from your independent work. Write the correct method and explain precisely why your original answer was wrong.

Next step: revisit the model or a relevant revision-guide section if needed.